

Déploiement

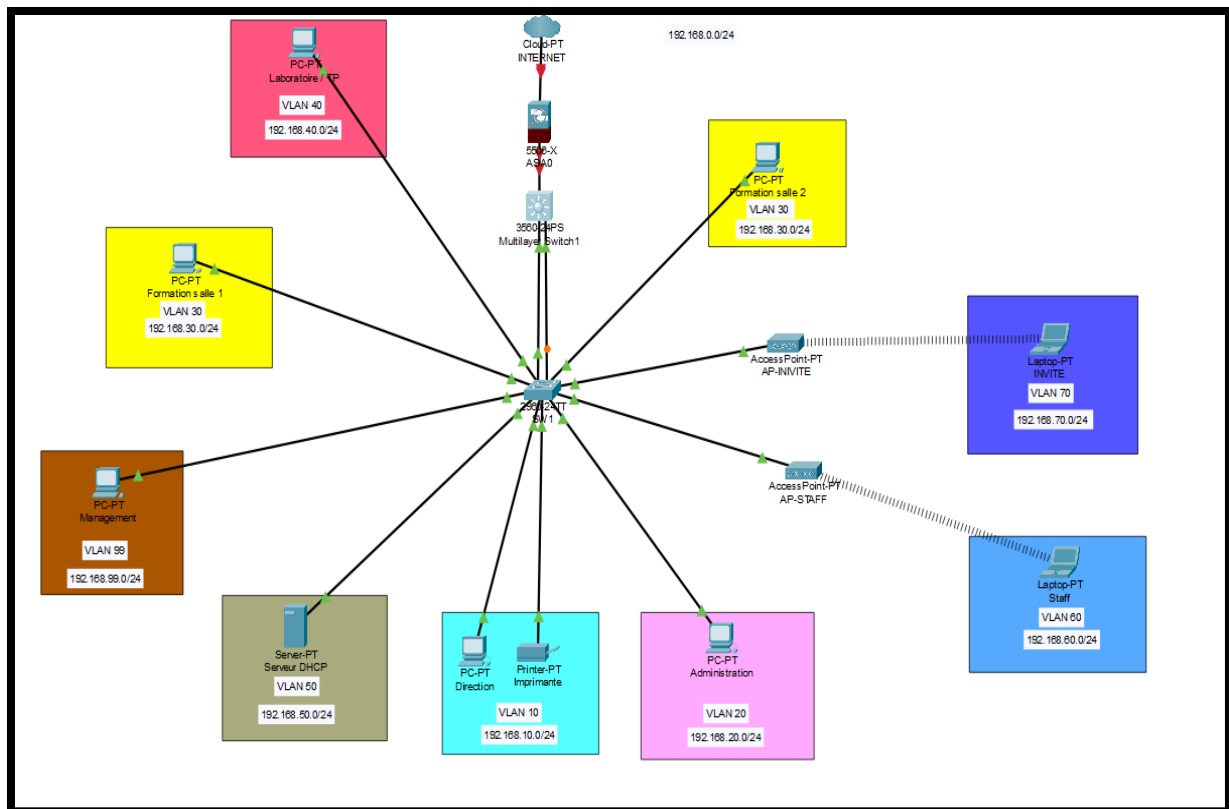


Figure 1 : Ajout des équipements et leur branchement

Étape 0 — Préparation du projet

1. Ouvrir Packet Tracer.
2. Créer un nouveau projet (File → New).
3. Ajouter les équipements suivants dans la zone de travail :
 - 1 × Switch 3560 (SW1-CORE)
 - 1 × Switch 2960 (SW2-ACCESS)
 - 1 × Cisco ASA 5505
 - 1 × Cloud-PT
 - 8 × PC-PT (postes clients)
 - 1 × Server-PT (SRV-DHCP)
 - Une imprimante

Étape 1 : Câblage (à réaliser en premier)

- Liaison SW1-CORE ↔ SW2-ACCESS (EtherChannel)
- Liaison SW1-CORE ↔ ASA
- Liaison ASA ↔ Cloud-PT

Branchement des postes sur SW2-ACCESS

Tableau 1 : Présentation des ports

Port SW2-ACCESS	Poste	VLAN
Fa0/1	PC-DIRECTION	10
Fa0/2	PC-ADMIN	20
Fa0/3	PC-FORMATION1	30
Fa0/4	PC-LABO	40
Fa0/5	SRV-DHCP	50
Fa0/6	PC-WIFI-STAFF	60
Fa0/7	PC-WIFI-GUEST	70
Fa0/11	PC-FORMATION2	80
Fa0/9	PC-MANAGEMENT	99

Étape 2 — Configuration de SW1-CORE

```
! Connexion et passage en mode configuration
Switch> enable
Switch# configure terminal
! Nom de l'équipement
hostname SW1-CORE
! Création des VLAN
vlan 10
  name DIRECTION
exit
vlan 20
  name ADMINISTRATION
exit
vlan 30
  name FORMATION1
exit
vlan 40
  name LABORATOIRE
exit
vlan 50
  name SERVEUR-DHCP
exit
vlan 60
  name WIFI-STAFF
exit
vlan 70
  name WIFI-GUEST
exit
vlan 80
```

```
name FORMATION2
exit
vlan 99
name MANAGEMENT
exit

! EtherChannel (LACP)
interface range gigabitEthernet0/1-2
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 1 mode active
exit

interface port-channel 1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
exit

! Activation du routage inter-VLAN
ip routing

! Interfaces SVI (passerelles de chaque VLAN)
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
ip helper-address 192.168.50.10
exit

interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
ip helper-address 192.168.50.10
exit

interface vlan 30
ip address 192.168.30.1 255.255.255.0
no shutdown
ip helper-address 192.168.50.10
exit

interface vlan 40
ip address 192.168.40.1 255.255.255.0
no shutdown
ip helper-address 192.168.50.10
exit

interface vlan 50
ip address 192.168.50.1 255.255.255.0
```

```
no shutdown
exit
```

```
interface vlan 60
ip address 192.168.60.1 255.255.255.0
no shutdown
ip helper-address 192.168.50.10
exit
```

```
interface vlan 70
ip address 192.168.70.1 255.255.255.0
no shutdown
ip helper-address 192.168.50.10
exit
```

```
interface vlan 80
ip address 192.168.80.1 255.255.255.0
no shutdown
ip helper-address 192.168.50.10
exit
```

```
interface vlan 99
ip address 192.168.99.1 255.255.255.0
no shutdown
exit
```

```
! Route par défaut vers l'ASA
ip route 0.0.0.0 0.0.0.0 10.255.255.1
```

```
! Configuration SSH
ip domain-name skillgate.local
username admin privilege 15 secret Cisco@123
crypto key generate rsa
! Choisir une taille de clé de 1024 bits
ip ssh version 2
line vty 0 15
transport input ssh
login local
exit
```

```
*****
```

```
Configuration du port fa0/24 pour la liaison de sécurité à ASA
```

```
enable
configure terminal
interface FastEthernet0/24
no switchport
ip address 10.255.255.2 255.255.255.252
no shutdown
exit
ip route 0.0.0.0 0.0.0.0 10.255.255.1
```

```
end
write memory
! Sauvegarde de la configuration
copy running-config startup-config
```

Étape 3 — Configuration de SW2-ACCESS

```
Switch> enable
Switch# configure terminal
hostname SW2-ACCESS
! Création des VLAN (identique à SW1-CORE)
vlan 10
  name DIRECTION
exit
vlan 20
  name ADMINISTRATION
exit
vlan 30
  name FORMATION1
exit
vlan 40
  name LABORATOIRE
exit
vlan 50
  name SERVEUR-DHCP
exit
vlan 60
  name WIFI-STAFF
exit
vlan 70
  name WIFI-GUEST
exit
vlan 80
  name FORMATION2
exit
vlan 99
  name MANAGEMENT
exit

! EtherChannel (LACP)
interface range gigabitEthernet0/1-2
  switchport trunk encapsulation dot1q
  switchport mode trunk
  channel-group 1 mode active
exit

interface port-channel 1
  switchport trunk encapsulation dot1q
  switchport mode trunk
```

```
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
exit
```

! Configuration des ports d'accès (postes)

```
interface range fastEthernet 0/1-2
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
spanning-tree portfast
```

```
exit
```

```
interface range fastEthernet 0/3-4
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
spanning-tree portfast
```

```
exit
```

```
interface range fastEthernet 0/5-10
```

```
switchport mode access
```

```
switchport access vlan 30
```

```
spanning-tree portfast
```

```
exit
```

```
interface range fastEthernet 0/11-15
```

```
switchport mode access
```

```
switchport access vlan 80
```

```
spanning-tree portfast
```

```
exit
```

```
interface range fastEthernet 0/16-18
```

```
switchport mode access
```

```
switchport access vlan 40
```

```
spanning-tree portfast
```

```
exit
```

```
interface fastEthernet 0/19
```

```
switchport mode access
```

```
switchport access vlan 50
```

```
exit
```

```
interface fastEthernet 0/20
```

```
switchport mode access
```

```
switchport access vlan 60
```

```
spanning-tree portfast
```

```
exit
```

```
interface fastEthernet 0/21
```

```
switchport mode access
```

```
switchport access vlan 70
```

```
spanning-tree portfast
```

```
exit
```

```
interface fastEthernet 0/22
```

```
switchport mode access
```

```
switchport access vlan 99
```

```
spanning-tree portfast
exit
interface range fastEthernet 0/23-24
switchport mode access
switchport access vlan 999
shutdown
exit
! Passerelle par défaut pour le management
ip default-gateway 192.168.99.1
! Configuration SSH
ip domain-name skillgate.local
username admin privilege 15 secret Cisco@123
crypto key generate rsa
! Choisir une taille de clé de 1024 bits
ip ssh version 2
line vty 0 15
transport input ssh
login local
exit
! Sauvegarde de la configuration
copy running-config startup-config
```

Étape 4 — Configuration de l'ASA

Le pare-feu FW-ASA assure la séparation entre le réseau interne de SkillGate et le réseau externe. Il réalise notamment le routage vers Internet, le NAT/PAT et l'administration sécurisée par SSH.

```
enable
! Passe du mode utilisateur au mode privilégié.
configure terminal
! Passe en mode de configuration globale.
hostname FW-ASA
! Définit le nom du pare-feu.
```

INTERFACE OUTSIDE : CONNEXION VERS INTERNET / CLOUD

```
interface GigabitEthernet1/1
! Sélectionne l'interface physique connectée au réseau externe.
nameif outside
! Donne le nom logique "outside" à cette interface.
security-level 0
! Attribue le niveau de sécurité le plus faible à l'interface externe.
ip address 203.0.113.2 255.255.255.252
! Attribue l'adresse IP 203.0.113.2/30 à l'interface externe.
no shutdown
! Active l'interface.
exit
```

! Quitte la configuration de l'interface.

```
FW-ASA(config)#interface gigabitEthernet 1/1
FW-ASA(config-if)#nameif outside
FW-ASA(config-if)#security-level 0
FW-ASA(config-if)#ip address 203.0.113.2 255.255.255.252
FW-ASA(config-if)#no shu
FW-ASA(config-if)#no shutdown
FW-ASA(config-if)#
FW-ASA(config-if)#do wr
FW-ASA(config)#
FW-ASA(config)#exi
FW-ASA(config)#exit
FW-ASA#wr
% Incomplete command.
FW-ASA#
```

Figure 2 : configuration de base du pare-feu ASA

INTERFACE INSIDE : CONNEXION VERS SW1-CORE

interface GigabitEthernet1/2

! Sélectionne l'interface connectée au switch central SW1-CORE.

nameif inside

! Donne le nom logique "inside" à cette interface.

security-level 100

! Attribue le niveau de sécurité le plus élevé au réseau interne.

ip address 10.255.255.1 255.255.255.252

! Attribue l'adresse IP 10.255.255.1/30 à l'interface interne.

! SW1-CORE utilise 10.255.255.2 sur le même lien.

no shutdown

! Active l'interface.

exit

! Quitte la configuration de l'interface.

```
FW-ASA(config)#in
FW-ASA(config)#interface g
FW-ASA(config)#interface gigabitEthernet 1/2
FW-ASA(config-if)#nameif inside
INFO: Security level for "inside" set to 100 by default.
FW-ASA(config-if)#nameif inside
FW-ASA(config-if)#security-level 100
FW-ASA(config-if)#ip address 10.255.255.1 255.255.255.252
Waiting for the earlier webvpn instance to terminate...
Previous instance shut down.Starting a new one
FW-ASA(config-if)#no sh

FW-ASA(config-if)#exi
FW-ASA(config-if)#exit
FW-ASA(config)#
FW-ASA(config)#
```

Figure 3 : configuration de l'interface inside

NAT/PAT : ACCÈS DES RÉSEAUX INTERNES VERS INTERNET

object network LAN-SKILLGATE

! Crée un objet représentant les réseaux internes de SkillGate.

subnet 192.168.0.0 255.255.0.0

! Définit le réseau privé 192.168.0.0/16 comme réseau concerné par le NAT.

nat (inside,outside) dynamic interface

! Active-le NAT/PAT.

! Les adresses privées des utilisateurs sont traduites en utilisant

! l'adresse IP de l'interface outside du pare-feu.

exit

! Quitte la configuration de l'objet réseau.

```
FW-ASA(config)#object network LAN-SKILLGATE
FW-ASA(config-network-object)#subnet 192.168.0.0 255.255.0.0
FW-ASA(config-network-object)#nat (inside,outside) dynamic interface
FW-ASA(config-network-object)#exi
FW-ASA(config-network-object)#exit
FW-ASA#wr
% Incomplete command.
FW-ASA#
```

Figure 4 : configuration du NAT

route outside 0.0.0.0 0.0.0.0 203.0.113.1

! Indique au pare-feu que toute destination inconnue doit être

! envoyée vers 203.0.113.1, qui représente la passerelle externe.

ROUTE VERS LES RÉSEAUX INTERNES

route inside 192.168.0.0 255.255.0.0 10.255.255.2

! Indique que les réseaux internes 192.168.0.0/16 sont accessibles

! via SW1-CORE à l'adresse 10.255.255.2.

```
FW-ASA#conf t
FW-ASA(config)#route outside 0.0.0.0 0.0.0.0 203.0.113.1
FW-ASA(config)#route inside 192.168.0.0 255.255.0.0 10.255.255.2
FW-ASA(config)#exi
FW-ASA(config)#exit
FW-ASA#wri
FW-ASA#write m
FW-ASA#write memory
Building configuration...
Cryptochecksum: 6df92c28 286306b6 1f422c62 6e0e5c36
```

Figure 5 : configuration du outside

CONFIGURATION DE L'ACCÈS SSH

domain-name skillgate.local

! Définit le nom de domaine utilisé par l'équipement.

crypto key generate rsa modulus 1024

! Génère les clés RSA nécessaires au fonctionnement de SSH.

username admin password Cisco123

! Crée l'utilisateur administrateur "admin" avec le niveau de privilège 15.

aaa authentication ssh console LOCAL

! Configure l'ASA pour authentifier les utilisateurs SSH à partir

! de sa base d'utilisateurs locale.

ssh 192.168.0.0 255.255.0.0 inside

! Autorise les équipements du réseau interne à établir une connexion

! SSH vers le pare-feu via l'interface inside.

ssh timeout 10

! Définit à 10 minutes le délai d'expiration d'une session SSH inactive.

ssh version 2

! Force l'utilisation de SSH version 2, plus sécurisée que SSH version 1.

! SAUVEGARDE

write memory

! Sauvegarde la configuration actuelle dans la mémoire permanente.

```
FW-ASA(config)#
FW-ASA(config)#domain-name skillgate.local
FW-ASA(config)#crypto key generate rsa modulus 1024
WARNING: You have a RSA keypair already defined named <Default-RSA-Key>.

Do you really want to replace them? [yes/no]: y
Keypair generation process begin. Please wait...
FW-ASA(config)#username admin password Cisco123 privilege 15
                                     ^
% Invalid input detected at '^' marker.

FW-ASA(config)#username admin password Cisco123 privilege 15
                                     ^
% Invalid input detected at '^' marker.

FW-ASA(config)#username admin password Cisco123
FW-ASA(config)#aaa authentication ssh console LOCAL
FW-ASA(config)#ssh 192.168.0.0 255.255.0.0 inside
FW-ASA(config)#ssh timeout 10
FW-ASA(config)#ssh version 2
                                     ^
% Invalid input detected at '^' marker.

FW-ASA(config)#end
FW-ASA#wri
FW-ASA#write me
FW-ASA#write memory
Building configuration...
Cryptochecksum: 6df92c28 286306b6 1f422c62 6e0e5c36

1478 bytes copied in 2.295 secs (644 bytes/sec)
[OK]
FW-ASA#
```

Figure 6 : Configuration de l'accès ssh

Vérification des interfaces

```
FW-ASA#show interface ip brief
Interface      IP-Address      OK? Method Status      Protocol
Virtual0       127.1.0.1       YES unset    up          up
GigabitEthernet1/1  203.0.113.2    YES manual  up          up
GigabitEthernet1/2  10.255.255.1    YES manual  up          up
GigabitEthernet1/3  unassigned      YES unset    administratively down down
GigabitEthernet1/4  unassigned      YES unset    administratively down down
GigabitEthernet1/5  unassigned      YES unset    administratively down down
GigabitEthernet1/6  unassigned      YES unset    administratively down down
GigabitEthernet1/7  unassigned      YES unset    administratively down down
GigabitEthernet1/8  unassigned      YES unset    administratively down down
Management1/1      unassigned      YES unset    administratively down down
Internal-Controll1/1  127.0.1.1     YES unset    up          up
Internal-Data1/1    unassigned      YES unset    up          up
Internal-Data1/2    unassigned      YES unset    up          up
Internal-Data1/3    unassigned      YES unset    up          up
FW-ASA#
```

Figure 7 : Vérification des interfaces

Test le plus important : ASA → SW1-CORE

ping 10.255.255.2

Test sur la passerelle

ping 203.0.113.1

```
FW-ASA#ping 10.255.255.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.255.255.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

FW-ASA#ping 203.0.113.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

FW-ASA#
```

Figure 8 : test de ASA vers SW1 - CORE

Donc le lien SW1-CORE ↔ ASA ↔ Cloud est opérationnel

Vérification des ensuite des routes

```
FW-ASA#show route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

    10.0.0.0/30 is subnetted, 2 subnets
C      10.0.0.0 255.255.255.252 is directly connected, inside, GigabitEthernet1/2
C      10.255.255.0 255.255.255.252 is directly connected, inside, GigabitEthernet1/2
S      192.168.0.0 255.255.0.0 [1/0] via 10.255.255.2, inside
    203.0.113.0/30 is subnetted, 1 subnets
C      203.0.113.0 255.255.255.252 is directly connected, outside, GigabitEthernet1/1
S*     0.0.0.0/0 [1/0] via 203.0.113.1
FW-ASA#
FW-ASA#
```

Figure 9 : Vérification des routes

Vérifie le NAT

```
FW-ASA#show nat
Auto NAT Policies (Section 2)
1 (inside) to (outside) source dynamic LAN-SKILLGATE interface
  translate_hits = 0, untranslate_hits = 0

FW-ASA#
FW-ASA#
FW-ASA#
```

Figure 10 : Vérification de NAT

Communication autorisée : PC interne vers Internet

Transformation d'un routeur en internet

```
enable
configure terminal
interface g0/0
ip address 203.0.113.1 255.255.255.0
no shutdown
exit
ip dhcp excluded-address 203.0.113.1 203.0.113.10
ip dhcp pool INTERNET
network 203.0.113.0 255.255.255.0
default-router 203.0.113.1
dns-server 8.8.8.8
exit
```

Pour le ASA puisse avoir des adresses

```
enable
configure terminal
interface gigabitEthernet 1/1
nameif outside
security-level 0
ip address dhcp
no shutdown
exit
```

write memory

4. Test final PC → Internet simulé

Comme notre objectif est de faire le **test de ping PC → Internet simulé**, ajoutons l'inspection ICMP..

Sur l'ASA

Entrons exactement :

Configure terminal

policy-map global_policy

class inspection_default

inspect icmp

exit

write memory

Depuis le même PC :

ping 198.51.100.8

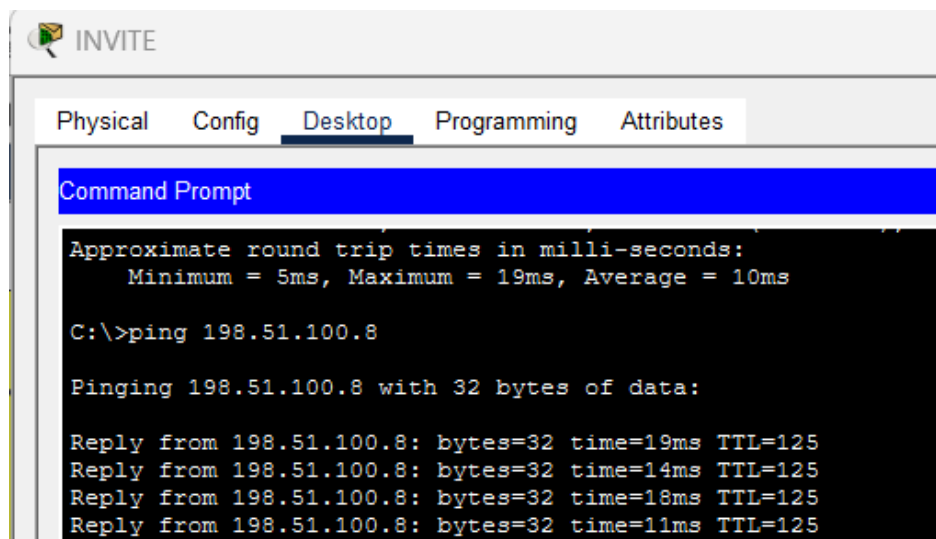


Figure 11 : pc vers internet

Étape 5 — Configuration du serveur DHCP

Adressage IP fixe du serveur

Cliquer sur SRV-DHCP.

Onglet Desktop → IP Configuration.

Renseigner :

Adresse IP : 192.168.50.10

Masque de sous-réseau : 255.255.255.0

Passerelle par défaut : 192.168.50.1

Serveur DNS : 8.8.8.8

Création des pools DHCP

Onglet Services → DHCP, puis créer huit pools (bouton Add pour chacun) :

Tableau 2 : Création des Pools sur le serveur

Pool	VLAN	Passerelle	DNS	Adresse de départ	Masque	Max. utilisateurs
POOL-DIRECTION	10	192.168.10.1	8.8.8.8	192.168.10.10	255.255.255.0	50
POOL-ADMIN	20	192.168.20.1	8.8.8.8	192.168.20.10	255.255.255.0	50
POOL-FORMATION1	30	192.168.30.1	8.8.8.8	192.168.30.10	255.255.255.0	50
POOL-LABO	40	192.168.40.1	8.8.8.8	192.168.40.10	255.255.255.0	50
POOL-WIFI-STAFF	60	192.168.60.1	8.8.8.8	192.168.60.10	255.255.255.0	50
POOL-WIFI-GUEST	70	192.168.70.1	8.8.8.8	192.168.70.10	255.255.255.0	50
POOL-FORMATION2	80	192.168.80.1	8.8.8.8	192.168.80.10	255.255.255.0	50
POOL-MANAGEMENT (optionnel)	99	192.168.99.1	8.8.8.8	192.168.99.10	255.255.255.0	20

Pour chaque pool, cliquer sur ON afin de l'activer.

Étape 6 — Configuration des postes clients

Pour chaque poste, à réaliser individuellement :

- Cliquer sur le poste.
- Onglet Desktop → IP Configuration.
- Sélectionner DHCP.
- Attendre 10 à 15 secondes que l'adresse IP s'affiche.
- Noter l'adresse IP attribuée.

Étape 7 — Vérifications et tests

Vérifications sur SW1-CORE

```
show vlan brief
show etherchannel summary
show ip interface brief
show ip route
```

```
SW1-CORE#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
10	DIRECTION	active	
20	ADMINISTRATION	active	
30	FORMATION1	active	
40	LABORATOIRE	active	
50	SERVEUR-DHCP	active	
60	WIFI-STAFF	active	
70	WIFI-GUEST	active	
80	FORMATION2	active	
99	MANAGEMENT	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW1-CORE#
```

Figure 12 : Vérification de la création des VLANS sur SW1

```
SW1-CORE#show etherchannel summary
```

Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port

Number of channel-groups in use: 1
Number of aggregators: 1

Group	Port-channel	Protocol	Ports
1	Po1(SU)	LACP	Gig0/1(P) Gig0/2(P)

```
SW1-CORE#
```

Figure 13 : Vérification de Etherchannel sur SW1

```
SW1-CORE#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Port-channel1  unassigned      YES unset  up          up
FastEthernet0/1 unassigned      YES unset  down        down
FastEthernet0/2 unassigned      YES unset  down        down
FastEthernet0/3 unassigned      YES unset  down        down
FastEthernet0/4 unassigned      YES unset  down        down
FastEthernet0/5 unassigned      YES unset  down        down
FastEthernet0/6 unassigned      YES unset  down        down
FastEthernet0/7 unassigned      YES unset  down        down
FastEthernet0/8 unassigned      YES unset  down        down
FastEthernet0/9 unassigned      YES unset  down        down
FastEthernet0/10 unassigned      YES unset  down        down
FastEthernet0/11 unassigned      YES unset  down        down
FastEthernet0/12 unassigned      YES unset  down        down
FastEthernet0/13 unassigned      YES unset  down        down
FastEthernet0/14 unassigned      YES unset  down        down
FastEthernet0/15 unassigned      YES unset  down        down
FastEthernet0/16 unassigned      YES unset  down        down
FastEthernet0/17 unassigned      YES unset  down        down
FastEthernet0/18 unassigned      YES unset  down        down
FastEthernet0/19 unassigned      YES unset  down        down
FastEthernet0/20 unassigned      YES unset  down        down
FastEthernet0/21 unassigned      YES unset  down        down
FastEthernet0/22 unassigned      YES unset  down        down
FastEthernet0/23 unassigned      YES unset  down        down
FastEthernet0/24 unassigned      YES unset  down        down
GigabitEthernet0/1 unassigned      YES unset  up          up
GigabitEthernet0/2 unassigned      YES unset  up          up
Vlan1          unassigned      YES unset  administratively down down
Vlan10         192.168.10.1    YES manual up          up
Vlan20         192.168.20.1    YES manual up          up
Vlan30         192.168.30.1    YES manual up          up
Vlan40         192.168.40.1    YES manual up          up
Vlan50         192.168.50.1    YES manual up          up
Vlan60         192.168.60.1    YES manual up          up
Vlan70         192.168.70.1    YES manual up          up
Vlan80         192.168.80.1    YES manual up          up
Vlan99         192.168.99.1    YES manual up          up
SW1-CORE#
```

Figure 14 : vérification du vlan 99 sur SW1

```
SW1-CORE# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
C    192.168.30.0/24 is directly connected, Vlan30
C    192.168.40.0/24 is directly connected, Vlan40
C    192.168.50.0/24 is directly connected, Vlan50
C    192.168.60.0/24 is directly connected, Vlan60
C    192.168.70.0/24 is directly connected, Vlan70
C    192.168.80.0/24 is directly connected, Vlan80
C    192.168.99.0/24 is directly connected, Vlan99

SW1-CORE#
```

Figure 15 : vérification des routes sur SW1

Vérifications sur SW2-ACCESS

```
show vlan brief
show etherchannel summary
show interfaces status
```

```
SW2-ACCESS#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2
10   DIRECTION                active    Fa0/3, Fa0/4
20   ADMINISTRATION          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
30   FORMATION1              active    Fa0/9, Fa0/10
40   LABORATOIRE             active    Fa0/16, Fa0/17, Fa0/18
50   SERVEUR-DHCP            active    Fa0/19
60   WIFI-STAFF              active    Fa0/20
70   WIFI-GUEST              active    Fa0/21
80   FORMATION2              active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
99   MANAGEMENT              active    Fa0/22
999  VLAN0999                active    Fa0/23, Fa0/24
1002 fddi-default            active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active
SW2-ACCESS#
```

Figure 16 : vérification de création des VLANS sur SW2

```
SW2-ACCESS#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SU)      LACP        Gig0/1 (P) Gig0/2 (P)
SW2-ACCESS#
```

Figure 17 : Vérification de etherchannel sur SW2

```
SW2-ACCESS#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Pol		connected	trunk	a-full	a-100	
Fa0/1		connected	10	a-full	a-100	10/100BaseTX
Fa0/2		connected	10	a-full	a-100	10/100BaseTX
Fa0/3		connected	20	a-full	a-100	10/100BaseTX
Fa0/4		notconnect	20	auto	auto	10/100BaseTX
Fa0/5		connected	30	a-full	a-100	10/100BaseTX
Fa0/6		notconnect	30	auto	auto	10/100BaseTX
Fa0/7		notconnect	30	auto	auto	10/100BaseTX
Fa0/8		notconnect	30	auto	auto	10/100BaseTX
Fa0/9		notconnect	30	auto	auto	10/100BaseTX
Fa0/10		notconnect	30	auto	auto	10/100BaseTX
Fa0/11		connected	80	a-full	a-100	10/100BaseTX
Fa0/12		notconnect	80	auto	auto	10/100BaseTX
Fa0/13		notconnect	80	auto	auto	10/100BaseTX
Fa0/14		notconnect	80	auto	auto	10/100BaseTX
Fa0/15		notconnect	80	auto	auto	10/100BaseTX
Fa0/16		connected	40	a-full	a-100	10/100BaseTX
Fa0/17		notconnect	40	auto	auto	10/100BaseTX
Fa0/18		notconnect	40	auto	auto	10/100BaseTX
Fa0/19		connected	50	a-full	a-100	10/100BaseTX
Fa0/20		connected	60	a-full	a-100	10/100BaseTX
Fa0/21		connected	70	a-full	a-100	10/100BaseTX
Fa0/22		connected	99	a-full	a-100	10/100BaseTX
Fa0/23		disabled	999	auto	auto	10/100BaseTX
Fa0/24		disabled	999	auto	auto	10/100BaseTX
Gig0/1		connected	trunk	a-full	a-100	10/100/1000BaseTX
Gig0/2		connected	trunk	a-full	a-100	10/100/1000BaseTX

```
SW2-ACCESS#
```

Figure 18 : Vérification des interfaces sur SW2

7.3 Vérifications sur l'ASA

```
show interface ip brief
show nat
show xlate
```

7.4 Tests de connectivité (depuis chaque poste)

Ouvrir l'invite de commandes sur un poste et exécuter :

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

Wireless0 Connection:(default port)

    Connection-specific DNS Suffix.:
    Link-local IPv6 Address.....: FE80::2D0:D3FF:FEEB:5530
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.70.10
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                        192.168.70.1

Bluetooth Connection:

    Connection-specific DNS Suffix.:
    Link-local IPv6 Address.....: ::
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                        0.0.0.0

C:\>|
```

Figure 19 : Vérification du fonctionnement du DHCP

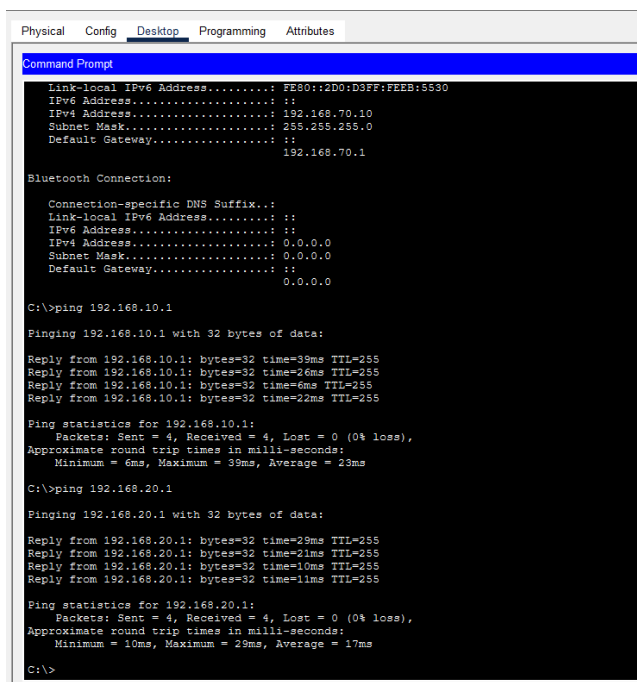
! 2. Tester la passerelle du VLAN concerné
ping 192.168.10.1

! 3. Tester un autre VLAN
ping 192.168.20.1

! 4. Tester l'ASA (interface inside)
ping 10.255.255.1

! 5. Tester l'ASA (interface outside)
ping 203.0.113.2

! 6. Tester Internet (Cloud-PT)
ping 203.0.113.1



```
Physical  Config  Desktop  Programming  Attributes
Command Prompt
Link-local IPv6 Address . . . . . : FE80::2D0:D3FF:FE5B:5530
IPv6 Address . . . . . : ::
IPv4 Address . . . . . : 192.168.70.10
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : ::
192.168.70.1

Bluetooth Connection:
Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : 
IPv6 Address . . . . . : ::
IPv4 Address . . . . . : 0.0.0.0
Subnet Mask . . . . . : 0.0.0.0
Default Gateway . . . . . : 
0.0.0.0

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=39ms TTL=255
Reply from 192.168.10.1: bytes=32 time=26ms TTL=255
Reply from 192.168.10.1: bytes=32 time=6ms TTL=255
Reply from 192.168.10.1: bytes=32 time=22ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 39ms, Average = 23ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=29ms TTL=255
Reply from 192.168.20.1: bytes=32 time=21ms TTL=255
Reply from 192.168.20.1: bytes=32 time=10ms TTL=255
Reply from 192.168.20.1: bytes=32 time=11ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 29ms, Average = 17ms

C:\>
```

Figure 20 : Test de connectivité entre les pc

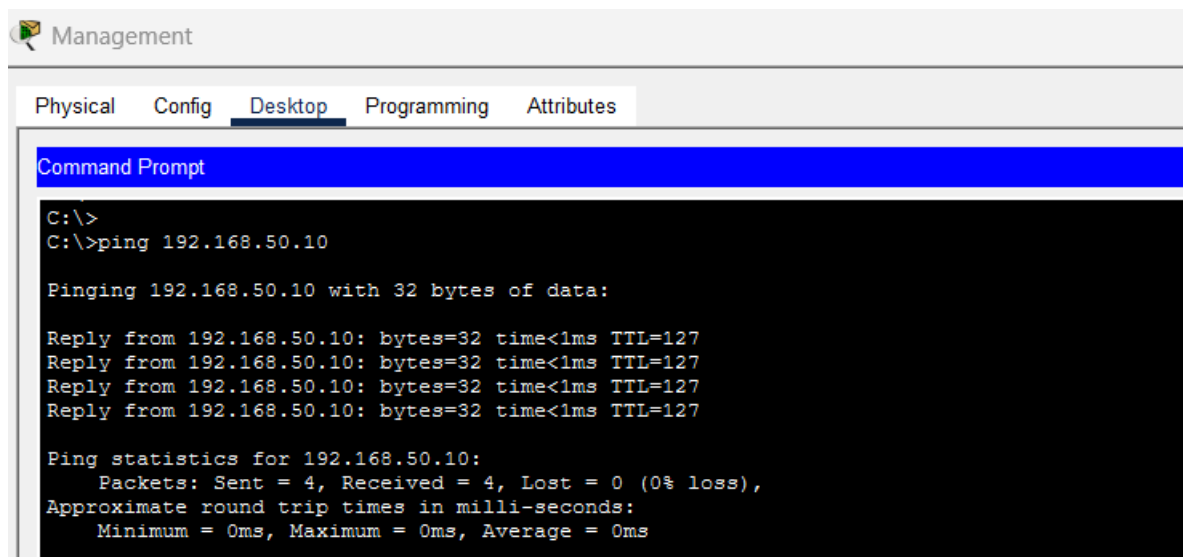


Figure 21 : test de connectivité entre les pc

7.5 Test SSH (depuis PC-MANAGEMENT)

Cliquer sur PC-MANAGEMENT.

Onglet Desktop → Command Prompt.

Saisir :

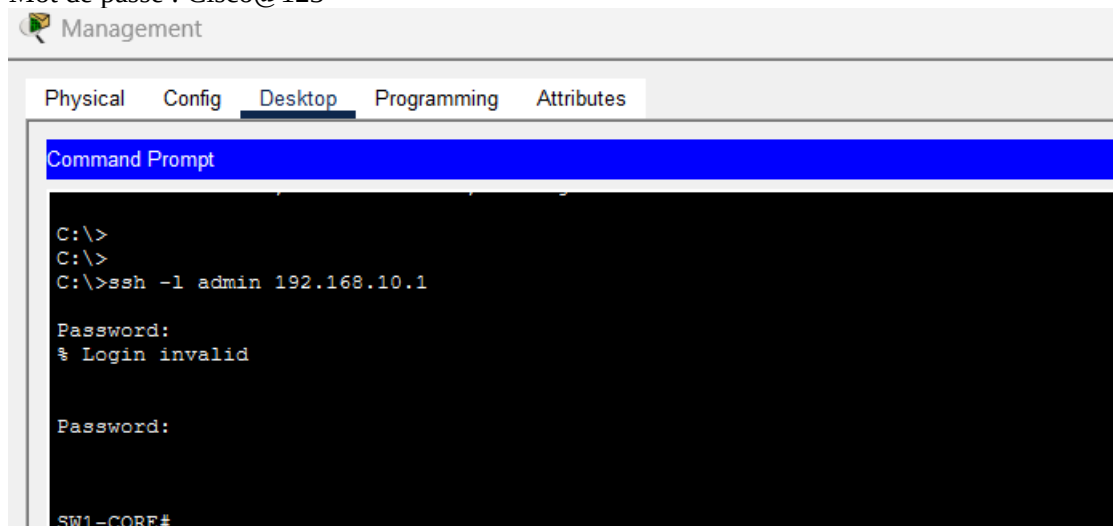
```
ssh -l admin 192.168.10.1
```

Mot de passe : Cisco@123

4. Répéter le test sur SW2-ACCESS :

```
ssh -l admin 192.168.99.2
```

Mot de passe : Cisco@123



Conclusion

Cette mise en œuvre a permis de démontrer, sur un périmètre volontairement limité aux éléments essentiels, la faisabilité de l'architecture proposée pour SkillGate Group : une segmentation logique par VLAN, une distribution automatique d'adresses IP centralisée, une agrégation de liens redondante entre les commutateurs, un premier niveau de sécurité périmétrique via le pare-feu ASA, et un accès d'administration chiffré par SSH. Cette base

pourrait être enrichie, dans une perspective d'évolution du projet, par des mécanismes de sécurité complémentaires (ACL plus fines, VLAN natif dédié, authentification 802.1X, journalisation centralisée).